UNIVERSITI TEKNOLOGI MALAYSIA

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SECJ3104 APPLICATION DEVELOPMENT

SECTION 02

PROJECT PROPOSAL

UTM BORROW

A Campus-Wide Circular Resource Sharing Platform

Prepared For	**DR. MOHD YAZID BIN IDRIS**
Submission Date	_____ OCTOBER 2025
Group Name	[Your Group Name]

NAME	MATRIC NUMBER
[Member 1 Full Name]	A23CSXXXX
[Member 2 Full Name]	A23CSXXXX
[Member 3 Full Name]	A23CSXXXX
[Member 4 Full Name]	A23CSXXXX

TABLE OF CONTENTS

1.0 Executive Summary 3
2.0 Project Domain and Community Background 4
3.0 Problem Analysis 5
3.1 Questionnaire Design and Distribution 5
3.2 Demographic Information 5
3.3 Resource Access and Financial Burden 6
3.4 Borrowing Behaviour and Platform Preferences 7
3.5 Trust and Accountability Concerns 8
3.6 Feature Desirability 9
4.0 Problem Statement 10
5.0 Goal and Objectives 11
6.0 Proposed Application 12
6.1 User Management and Authentication Subsystem 13
6.2 Inventory and Listing Subsystem 14
6.3 Borrow Request and QR Handover Subsystem 15
6.4 Analytics and Administration Subsystem 16
7.0 Proposed Technology and Tools Used 17
8.0 Project Benefits 19
9.0 Project Schedule — Waterfall Phases 20
10.0 Team Members and Roles 21
11.0 Individual Schedule 22

1.0 EXECUTIVE SUMMARY

UTM Borrow is a proposed mobile application designed to build a secure, authenticated, and highly efficient digital resource-sharing ecosystem exclusively for the Universiti Teknologi Malaysia (UTM) community. The platform's primary function is to facilitate the peer-to-peer (P2P) borrowing and lending of physical academic resources — including engineering instruments, laboratory equipment, specific course textbooks, architectural tools, and electronics — among verified UTM students and staff.

The initiative is driven by a critical economic inefficiency observed across the campus: students frequently purchase expensive, module-specific items for one-time use only to have them become dormant assets in their dormitory rooms. This "buy once, discard" cycle creates unnecessary financial pressure on students, particularly those from lower-income backgrounds, while simultaneously generating preventable material waste that contradicts the university's sustainability goals.

Unlike the informal and unmoderated barter channels that currently exist in WhatsApp and Telegram groups — where posts are buried instantly, identities are unverified, and no accountability mechanism exists — UTM Borrow is engineered as an official, governance-enforced platform. It authenticates every user against the UTM student and staff portal, tracks item custody in real time through a Lease Cycle State Machine, and secures every physical handover via a time-limited cryptographic QR token that expires in 60 seconds to prevent screenshot-based fraud.

The proposed application architecture is segmented into four interconnected subsystems: User Management and Authentication, Inventory and Listing, Borrow Request and QR Handover, and Analytics and Administration. Development will follow the Waterfall software development methodology — proceeding sequentially through Requirements, System Design, Implementation, Testing, and Deployment phases — with all project tasks tracked in Jira, system diagrams produced in Visual Paradigm, UI/UX prototypes designed in Figma, and source code managed via GitHub. Frontend code will be built using React Native based on Figma-generated designs, supported by a Node.js/Firebase backend.

In the long term, UTM Borrow is designed to evolve from a zero-cost utility into a self-sustaining platform through a symbolic 1 RM maintenance fee per transaction, a Merit Point gamification engine that rewards high-volume lenders with university recognition, and a Business Intelligence dashboard that allows the UTM administration and UTMCS to quantify student savings and material waste diversion in real time.

2.0 PROJECT DOMAIN AND COMMUNITY BACKGROUND

Universiti Teknologi Malaysia (UTM) is Malaysia's leading engineering and technology university, home to over 20,000 students spanning undergraduate, postgraduate, and international programmes across multiple faculties including Computing, Mechanical Engineering, Electrical Engineering, Chemical Engineering, and Architecture. The diversity of academic programmes means that students are routinely required to acquire highly specialized, module-specific physical resources — from precision measuring instruments and lab-safe electronics to architectural drawing tools and rare postgraduate reference texts.

2.1 The Current Problem Space

At present, the UTM community has no official, centralized platform dedicated to the sharing or borrowing of academic physical resources. The current landscape relies entirely on informal digital channels — primarily residential college WhatsApp and Telegram groups — which suffer from four systemic failures:

- Unverified Identity: These open groups permit non-students to participate, creating a risk of scams and fraudulent transactions with no accountability mechanism.
- Zero Discoverability: Item requests are buried beneath continuous administrative messages within minutes of posting, making it practically impossible to match lenders and borrowers efficiently.
- No Accountability Layer: When items are damaged or not returned, there is no institutional enforcement pathway. Disputes devolve into interpersonal conflict with no resolution.
- No State Tracking: The platform has no way of knowing who currently holds an item, when it was borrowed, or when it is due back — creating a system built entirely on trust with no verification.

2.2 Community and Stakeholder Context

The UTM Borrow platform operates within the high-impact domain of campus sustainability and student financial welfare. Its core stakeholders are:

Stakeholder	Role in UTM Borrow	Primary Benefit
UTM Students	Primary users — both Lenders and Borrowers	Reduce cost of single-use academic purchases; earn Merit Points
UTM Academic Staff	Optional Lenders of faculty-specific equipment	Contribute to the campus circular economy
UTM Administration	Platform governance and oversight	Data on student savings and waste diversion
UTMCS (Campus Sustainability)	Analytics consumer and sustainability partner	Quantified sustainability metrics for reporting
UTM IT Department	OAuth2 SSO integration provider	Secure identity layer for the closed ecosystem

UTM Borrow aligns directly with UTM's Smart Campus Vision and its commitment to sustainable development, effectively converting dormant single-use assets into a living, shared campus inventory that reduces waste, lowers costs, and builds community trust.

3.0 PROBLEM ANALYSIS

3.1 Questionnaire Design and Distribution

To validate the need for UTM Borrow and gather empirical evidence of the problem, the team designed and distributed a structured questionnaire to UTM students via faculty WhatsApp groups, college group chats, and direct outreach through course representatives. The questionnaire was hosted on Google Forms and collected responses over a two-week period, gathering a total of [X] responses from students across multiple faculties and year groups.

The questionnaire was structured into six thematic sections: demographic information, resource access and financial burden, borrowing behaviour and platform preferences, trust and accountability concerns, communication preferences, and feature desirability. This structure was designed to build a complete picture of the current pain points and the appetite for a structured borrowing solution.

3.2 Demographic Information

Demographic Variable	Breakdown	Significance
Gender	Male: [X]% | Female: [X]%	Broad gender representation ensures platform design is universally relevant
Year of Study	Year 1: [X]% | Year 2: [X]% | Year 3: [X]% | Year 4+: [X]% | Postgrad: [X]%	Higher year students have accumulated more dormant equipment — core lender base
Faculty	Computing, Mechanical Eng., Electrical Eng., Architecture, Others	Faculty diversity confirms cross-discipline demand for varied item categories
Residential Status	On-Campus: [X]% | Off-Campus: [X]%	On-campus students interact daily with limited storage — high motivation to lend

[Insert Figure 3.1: Pie Chart — Year of Study Distribution]

[Insert Figure 3.2: Bar Chart — Faculty Distribution of Respondents]

3.3 Resource Access and Financial Burden

The most critical finding of the questionnaire concerns the direct financial impact of single-module academic purchases on UTM students. Respondents were asked to report their experience with buying items they used only once or twice during their studies.

[Insert Figure 3.3: Pie Chart — Frequency of Purchasing Items Used for Only One Module]

Results revealed that a significant majority of students ([X]%) reported purchasing items they used for only one module before the items became unused. The average estimated total expenditure on such single-use items per semester was reported as RM [X]–RM [X], representing a meaningful and recurring financial burden on the student population. This data provides a direct economic justification for the UTM Borrow platform as a financial relief mechanism.

[Insert Figure 3.4: Bar Chart — Average Money Spent on Single-Module Items Per Semester]

Furthermore, [X]% of respondents indicated that they would have borrowed these items instead of purchasing them if a secure, verified borrowing platform had been available — confirming the latent demand that UTM Borrow is positioned to fulfil.

3.4 Borrowing Behaviour and Platform Preferences

[Insert Figure 3.5: Pie Chart — Current Methods Used to Find or Lend Items]

When asked about their current approach to sourcing needed academic items, [X]% of respondents reported primarily relying on friends or classmates through informal messaging, while [X]% had attempted to use WhatsApp or Telegram groups. Only [X]% had used any form of structured marketplace. This confirms that while the intent to share exists, the infrastructure to facilitate it in an organized manner is entirely absent.

[Insert Figure 3.6: Bar Chart — Preferred Communication Method for Coordinating Borrows]

Regarding communication preferences, [X]% of respondents preferred an in-app messaging system for coordinating borrowing, strongly validating the inclusion of a structured in-app negotiation and approval flow rather than redirecting users to external chat applications.

3.5 Trust and Accountability Concerns

The questionnaire identified trust as the primary barrier to participation in any sharing system. When asked about the most important factor for using a campus borrowing platform, seller and borrower trustworthiness ranked highest at [X]% of responses — significantly outweighing price ([X]%) and convenience ([X]%).

[Insert Figure 3.7: Pie Chart — Most Important Factor When Participating in a Campus Borrow System]

Critically, [X]% of respondents reported that fear of item damage or non-return was the primary reason they would hesitate to lend items to others — even if they wanted to. This finding directly validates the design decision to implement the QR Handover System and Condition Photo Log as mandatory, non-bypassable components of the transaction flow. The system does not ask lenders to trust borrowers personally; it asks the system to enforce accountability on their behalf.

[Insert Figure 3.8: Bar Chart — Primary Reasons for Reluctance to Lend Items]

Additionally, [X]% of respondents strongly agreed that they would trust a borrowing platform more if all users were verified UTM students or staff — reinforcing the decision to use UTM OAuth2 SSO authentication as the non-negotiable identity gate for the entire platform.

3.6 Feature Desirability

[Insert Figure 3.9: Bar Chart — Most Desired Features in a Campus Borrowing Platform]

Desired Feature	% Respondents Selecting	UTM Borrow Implementation
Verified UTM student identity	[X]%	OAuth2 SSO authentication against UTM portal
Real-time item availability tracking	[X]%	Lease Cycle State Machine — In Store / In Use / Pending
Secure in-person handover verification	[X]%	Dual QR scan with 60-second cryptographic token
Damage and loss protection	[X]%	Pre/Post-Borrow photo log + Trust Rating enforcement
Proximity-based item search	[X]%	Geolocation-filtered search by college building
Rating system for users	[X]%	Trust Rating (0–5) with Good Standing enforcement
Deadline reminders for returns	[X]%	Cloud Function automated 48hr and 24hr push notifications
Merit/reward for lending	[X]%	Merit Point gamification engine (Phase 2)

4.0 PROBLEM STATEMENT

The development of UTM Borrow is driven by four interconnected, empirically validated problems currently experienced by the UTM community, each with a direct technical solution embedded in the platform's architecture.

Problem 1: Financial Drain from Single-Use Academic Purchases

UTM students are routinely forced to purchase expensive, module-specific physical resources — tools, instruments, textbooks, and equipment — that they use for only one semester or a single assignment. Survey data confirms that [X]% of students have purchased items used only once, representing an estimated average expenditure of RM [X]–RM [X] per student per semester on items that immediately become dormant. This cycle creates a compounding financial burden with no recovery mechanism.

Technical Solution: The Inventory and Listing Subsystem allows verified UTM users to register these dormant items as borrowable assets. The proximity-filtered search engine surfaces available items to nearby borrowers in real time, converting idle inventory into a campus-wide shared resource pool.

Problem 2: Absence of Trust Infrastructure in Existing Sharing Channels

While informal sharing happens through WhatsApp and Telegram groups, it is entirely trust-based with zero accountability. Non-students can participate, identities cannot be verified, and there is no enforcement pathway when items are damaged or not returned. Survey data shows that [X]% of students cite fear of item loss or damage as the primary reason they do not lend their items to others — a fear that cannot be resolved by a chat application alone.

Technical Solution: The User Management and Authentication Subsystem enforces UTM OAuth2 SSO for all account creation, ensuring only verified UTM affiliates can access the platform. The Borrow Request and QR Handover Subsystem implements a mandatory dual-scan handover protocol using cryptographically signed, 60-second tokens — creating an irrefutable digital record of custody transfer that removes the human trust bottleneck.

Problem 3: No Accountability Mechanism for Item Returns and Condition

When items are returned damaged or not returned at all through informal channels, there is no institutional resolution pathway. Disputes are purely interpersonal, and students have no recourse. This lack of governance is the primary reason resource sharing does not happen at scale despite widespread willingness.

Technical Solution: The QR Handover Subsystem mandates Pre-Borrow and Post-Borrow photo documentation at both handover events. The Lease Cycle State Machine automatically transitions accounts to Overdue status and suspends Borrowing Rights when return deadlines are missed. The Admin Moderation Panel allows designated university administrators to resolve disputes and export PDF-validated compliance reports for disciplinary proceedings when necessary.

Problem 4: Wasted Campus Resources and Environmental Impact

The aggregate effect of thousands of single-use purchases per semester is measurable material waste — packaging, manufacturing resources, and physical items that are discarded or stored indefinitely. This contradicts UTM's commitment to sustainability and its ambition to become a zero-waste Smart Campus.

Technical Solution: The Analytics and Administration Subsystem provides a real-time Business Intelligence dashboard for the UTM administration and UTMCS, visualizing cumulative student savings (RM) and estimated material waste diversion (kg) generated by the platform. This transforms UTM Borrow from a student utility into a quantified sustainability instrument aligned with the university's institutional goals.

5.0 GOAL AND OBJECTIVES

The overarching goal of this project is to design, develop, and deploy a secure, authenticated, and accountability-enforced peer-to-peer academic resource borrowing platform — UTM Borrow — that reduces the financial burden of single-use academic purchases for UTM students, promotes a circular campus economy, and provides the university administration with quantified sustainability insights.

Core Technical Objectives

#	Objective	Success Criteria	Phase
O1	Implement a closed, identity-verified user ecosystem through UTM OAuth2 SSO authentication, ensuring only verified UTM students and staff can access or transact on the platform.	100% of accounts linked to valid UTM matriculation numbers; no self-registration with external email permitted.	Phase 1
O2	Build a distributed, geolocation-aware Inventory and Listing Subsystem that allows any UTM user to register their borrowable items with photo documentation, condition ratings, and real-time availability states.	Users can create, publish, search, and discover listings within 2 seconds on standard campus network.	Phase 1
O3	Implement a cryptographically secured QR Handover System with 60-second token expiry that creates an irrefutable digital record of every custody transfer, eliminating the primary barrier to lending (fear of non-return).	Zero successful fraudulent handovers via screenshot replay; all state transitions atomic and logged.	Phase 1
O4	Develop a full Lease Cycle State Machine with automated Cloud Function reminders at 48hr and 24hr before return deadlines, and automated Overdue state transition with Borrowing Rights suspension.	Reminder notification delivery within 5 seconds of trigger; 100% automated state transitions without manual intervention.	Phase 1
O5	Establish a Trust Rating system (0.0–5.0) tied to Good Standing enforcement that deters poor borrowing behaviour and rewards responsible participation.	Good Standing threshold (3.0+) correctly blocks Borrowing Rights for non-compliant users.	Phase 1
O6	Provide an Admin Moderation Panel with full transaction log access, dispute resolution tools, user suspension capability, and PDF export of compliance reports.	Administrators can resolve any dispute and export a validated report within the platform in under 5 minutes.	Phase 1
O7	Integrate a Merit Point gamification engine and symbolic 1 RM maintenance fee digital wallet to ensure long-term platform sustainability and active community participation.	Merit Points correctly awarded per completed lending cycle; fee revenue covers monthly Firebase server costs.	Phase 2

O8	Deploy an Analytics and BI Dashboard for UTMCS and UTM administration visualizing total student savings (RM), active transactions, items lent, and estimated waste diversion (kg).	Dashboard updates in real time from Firestore data; accessible to administrators without requiring manual data export.	Phase 2

6.0 PROPOSED APPLICATION

UTM Borrow is a mobile application (iOS and Android via React Native) with a companion progressive web application (PWA) for browser access. It functions as an intelligent inventory management and accountability-enforced borrowing engine for the UTM campus community — not a simple classified ads board. The platform treats the entire campus as a single shared warehouse, tracks the real-time custody status of every registered item, and enforces governance at every stage of the transaction lifecycle.

The architecture is built around four interconnected subsystems, each handling a distinct phase of the item lifecycle. Every subsystem communicates through a central Node.js REST API deployed on Firebase Cloud Functions, with all persistent data stored in Google Firebase Firestore and media files in Firebase Cloud Storage.

Subsystem	Modules	Core Responsibility
SS1 — User Management & Authentication	• Authentication & SSO Module • Profile & Trust Module • Role-Based Access Module	Identity verification, profile management, Trust Rating calculation, Good Standing enforcement
SS2 — Inventory & Listing	• Item Listing Module • Geolocation Search Module • Category & Condition Module	Borrowable item registration, photo upload, proximity search, real-time availability state tracking
SS3 — Borrow Request & QR Handover	• Request Flow Module • QR Token Engine Module • Lease Cycle State Module • Condition & Dispute Module	End-to-end transaction lifecycle management from request to return, cryptographic handover, automated reminders, damage reporting
SS4 — Analytics & Administration	• Admin Moderation Module • Notification Engine Module • BI Dashboard Module • Compliance Export Module	Platform governance, push notifications, sustainability analytics, dispute resolution, PDF compliance reporting

[Insert Figure 6.1: Hierarchical Subsystem and Module Diagram — generated in Visual Paradigm]

[Insert Figure 6.2: Overall Use Case Diagram — generated in Visual Paradigm]

6.1 User Management and Authentication Subsystem (SS1)

This subsystem is the identity and trust foundation of UTM Borrow. It is the mandatory gateway through which every user must pass before interacting with any other subsystem. The subsystem enforces the closed-ecosystem principle — no anonymous accounts, no external email registration, and no access without a valid UTM identity.

6.1.1 Authentication and SSO Module

This module handles the complete OAuth2 Single Sign-On flow against the UTM student and staff portal. Upon first successful login, the module automatically retrieves the user's matriculation number, full name, role (student/staff), faculty, and college affiliation, creating a verified profile in Firestore. All subsequent sessions are authenticated via signed JWT tokens with 24-hour expiry.

6.1.2 Profile and Trust Module

This module maintains each user's public profile including: display name, college, faculty, member since date, cumulative Trust Rating (0.0–5.0 weighted average of received ratings), total items lent, total borrows completed, and current Good Standing status. Trust Ratings are automatically recalculated after every completed transaction. Accounts with ratings below 3.0 are flagged and Borrowing Rights suspended.

6.1.3 Role-Based Access Control Module

This module enforces three distinct access roles: Student/Staff User (full platform access), Platform Administrator (Admin Panel access with moderation tools), and System (automated Cloud Functions). All API endpoints validate the user's role from their JWT before executing any operation, preventing unauthorized cross-role data access.

6.2 Inventory and Listing Subsystem (SS2)

This subsystem is the platform's distributed campus inventory engine. It transforms the scattered, dormant academic resources of individual students and staff into a structured, searchable, and real-time shared catalog — effectively turning the UTM campus into a single shared warehouse.

6.2.1 Item Listing Module

Authenticated users can register items for borrowing by providing: item name, category (Lab Equipment, Books, Stationery, Electronics, Workshop Tools, Clothing/PPE, Others), condition rating (New/Good/Fair/Poor), a minimum of one high-resolution photo, maximum borrow duration (days), and visibility setting (Public or Faculty-Only). Each listing is assigned one of three real-time states: Available, Pending, or In Use.

6.2.2 Geolocation Search Module

This module powers the platform's item discovery experience. All search queries are filtered by the Firestore Availability index (Available state only by default) and then sorted by the proximity of the lender's college to the borrower's college using pre-stored GPS coordinates. Users can additionally filter by category, condition, and availability. Results are paginated at 20 items per page to optimize mobile data consumption.

6.2.3 Category and Condition Module

This module enforces structured metadata on all listings, enabling the AI-ready data foundation for future recommendation engines. It also manages the pre-borrow photo archive — storing high-resolution baseline condition photos that are later compared against post-return photos in the dispute resolution flow.

6.3 Borrow Request and QR Handover Subsystem (SS3)

This is the most technically complex and strategically critical subsystem — it is what differentiates UTM Borrow from every informal channel that has been tried before. It manages the complete lifecycle of every borrow transaction, from the initial request through to the final return, with every state transition enforced by the system rather than by personal trust.

6.3.1 Request Flow Module

A user in Good Standing can send a borrow request for any Available item, specifying a desired duration within the lender's maximum. The lender receives a real-time push notification and can Approve or Reject from their dashboard. Approval transitions the item to Pending and triggers deadline calculation. The module maintains a full request history including rejections and cancellations for audit purposes.

6.3.2 QR Token Engine Module

This module is the cryptographic backbone of the platform's trust system. When a lender approves a request and is ready to hand over the item, they generate a Dispatch QR code — a HMAC-SHA256 signed JSON Web Token containing the transaction ID, lender ID, borrower ID, and a Unix timestamp. This token automatically expires 60 seconds after generation, making screenshot-based relay attacks technically impossible. The borrower scans the QR using the in-app camera (external QR readers are not accepted). An identical process occurs for the Return QR at the end of the lease.

6.3.3 Lease Cycle State Module

Every transaction passes through the following states enforced by the system: Request Sent → Request Approved → Dispatched (In Use) → Return Initiated → Returned (Closed). Firebase Cloud Functions automatically transition overdue transactions to Overdue state, send 48hr and 24hr deadline reminders, and flag borrower accounts. All state transitions are atomic (Firestore transactions) and fully logged in the audit collection.

6.3.4 Condition and Dispute Module

During the Return QR scan, both parties complete a Condition Check: the Borrower uploads a Post-Return Photo and the Lender rates the condition as Unchanged, Minor Damage, or Major Damage. If damage is reported, the system immediately freezes the Borrower's account, creates a Dispute record linking both photos, and routes the case to the Admin queue. Administrators can rule in favour of either party, with the system automatically applying the corresponding penalty or clearance.

6.4 Analytics and Administration Subsystem (SS4)

This subsystem provides the platform's governance, visibility, and sustainability reporting capabilities. It serves two distinct audiences: platform administrators (who require moderation tools and compliance reports) and UTM administration/UTMCS (who require sustainability analytics and institutional data).

6.4.1 Admin Moderation Module

Role-restricted to Platform Administrators. Provides a real-time dashboard of: active listings count, active borrow transactions, overdue accounts, dispute queue, registered user count, and platform-wide trust rating distribution. Administrators can suspend, warn, or reinstate any user account with mandatory reason logging and a full audit trail.

6.4.2 Notification Engine Module

Firebase Cloud Messaging (FCM) integration delivering push notifications for all transaction events: borrow requests, approvals/rejections, handover confirmations, deadline reminders (48hr and 24hr), overdue alerts, dispute updates, and rating prompts. All notifications are simultaneously delivered to the user's UTM email as a secondary channel. Users can configure notification preferences per event type.

6.4.3 BI Dashboard Module (Phase 2)

A data visualization dashboard accessible to UTM administration and UTMCS displaying: total RM savings generated by the platform (estimated value of borrowed items × number of transactions), total kg of material waste diverted from purchase, most borrowed item categories, college-level participation rates, and semester-over-semester trend charts. Built using Chart.js on the frontend, powered by Firestore aggregation queries.

6.4.4 Compliance Export Module

Administrators can export any transaction's complete Lease Cycle history as a formatted PDF report, including timestamped state transitions, user details, item details, condition photos, and all associated metadata. This document serves as the formal evidence package for UTM disciplinary proceedings in cases of significant violation.

7.0 PROPOSED TECHNOLOGY AND TOOLS USED

7.1 Development Stack

Component	Technology / Tool	Purpose and Justification
Mobile Frontend	React Native (Expo)	Cross-platform iOS and Android application from a single JavaScript codebase. Expo simplifies testing via QR-code live preview without native SDK installation. Figma-generated UI designs are translated into React Native components.
Web Frontend (PWA)	React.js	Progressive Web Application for browser access. Shares the same component library and business logic as the mobile frontend, avoiding duplication.
UI Prototyping	Figma	All screens are designed and prototyped in Figma before implementation. Figma can generate basic front-end code snippets as a starting reference, but production code is authored manually in React Native for quality and control.
Styling	Tailwind CSS (via NativeWind)	Utility-first styling for consistent, responsive design across screen sizes without writing custom CSS.
Backend API	Node.js + Express.js on Firebase Cloud Functions	Serverless Node.js REST API. Scales automatically with campus demand spikes. All business logic, state machine transitions, and authentication validation live here.
Database	Firebase Firestore (NoSQL)	Real-time document store with offline sync support. Schema-flexible for the evolving item/transaction data model. Native Cloud Function triggers for automated state transitions.
Authentication	UTM OAuth2 SSO + Firebase Auth	Closes the ecosystem to verified UTM users only. Firebase Auth manages session tokens; UTM OAuth2 provides the identity assertion.
Media Storage	Firebase Cloud Storage	Secure storage for item photos and condition report images. Access controlled by Firebase Security Rules tied to user authentication.
Push Notifications	Firebase Cloud Messaging (FCM)	Zero-cost push delivery for transaction events and deadline reminders to both iOS and Android.
QR Security	HMAC-SHA256 signed JWT (60-second expiry)	Cryptographic token preventing screenshot replay attacks. Generated server-side and validated server-side — never trusted from client.

7.2 Project Management and Collaboration Tools

All project management and collaboration activities follow the prescribed toolset defined by Dr. Mohd Yazid:

Tool	Purpose in This Project

Jira	Primary project tracker for the Waterfall phase plan. Each Waterfall phase (Requirements, Design, Implementation, Testing, Deployment) is structured as a Jira Epic, with individual features broken into Jira Stories and Tasks. All task assignments, progress tracking, and sprint reviews are managed within Jira.
Trello	Visual Kanban board used as a daily team coordination layer alongside Jira. Columns represent work states: Backlog, In Progress, In Review, Done. Used during team stand-up meetings for quick status updates.
Visual Paradigm	All system design artifacts are produced in Visual Paradigm, including: Use Case Diagrams, Class Diagrams, Sequence Diagrams, Activity Diagrams, Entity-Relationship Diagrams, and the Lease Cycle State Machine diagram. All diagrams will be exported and embedded in the final SRS and Design documents.
Figma	UI/UX design and interactive prototype tool. All screens, user flows, component libraries, and design tokens are built in Figma. The prototype demonstrates the full Happy Path and Error Path user journeys for presentation to the Dr. and stakeholders.
GitHub	Version control and collaborative code repository. The team follows a feature-branch workflow: each subsystem is developed on a dedicated branch, merged to develop via Pull Request after peer review, and released to main at Waterfall phase milestones.

8.0 PROJECT BENEFITS

The successful deployment of UTM Borrow delivers quantifiable and institutional benefits across multiple stakeholder groups:

8.1 For UTM Students

- Direct Financial Relief: Students gain access to a verified inventory of borrowable academic items, eliminating the need for single-module purchases. The projected campus-wide savings model estimates RM 600,000+ in avoided purchases within the first academic year at 10% adoption.
- Reduced Physical Clutter: Students who lend their idle items reclaim dormitory storage space and avoid the environmental guilt of discarding functional academic tools.
- Merit Recognition: High-volume lenders earn university-recognized Merit Points — a tangible, academic record of their contribution to the campus community, convertible to privileges or recognition in Phase 2.
- Safe Transaction Experience: The QR handover system, Trust Rating enforcement, and Admin dispute resolution pathway mean students can borrow and lend without personal trust risk — the system enforces accountability on their behalf.

8.2 For UTM Administration and UTMCS

- Quantified Sustainability Metrics: The BI Dashboard provides real-time data on student savings (RM) and material waste diverted from purchase (estimated kg) — providing UTMCS with documented evidence of circular economy impact for sustainability rankings and reporting.
- Reduced Administrative Burden: Centralizing campus commerce into a governed platform removes off-topic posts from official communication channels and provides a clear institutional resolution pathway for any disputes — reducing the ad-hoc burden on college administrators.
- Smart Campus Alignment: UTM Borrow directly advances the UTM Smart Campus Vision by deploying a data-driven, technology-integrated solution to a real and measurable campus-wide problem.

8.3 For the University as a Whole

- Environmental Impact: Every completed borrow transaction represents one fewer new item manufactured, packaged, and eventually discarded. At scale, this contributes meaningfully to UTM's zero-waste campus ambition.
- Student Entrepreneurship and Community: The platform builds a culture of trust, sharing, and community responsibility — aligning with UTM's mission to develop not just technical graduates but responsible, community-minded professionals.

9.0 PROJECT SCHEDULE — WATERFALL PHASES

UTM Borrow is developed following the Waterfall software development methodology, which enforces a sequential, phase-gated progression through clearly defined stages. Each phase must be formally completed and reviewed before the next phase begins, ensuring comprehensive documentation and design quality before a single line of production code is written.

Phase	Activities	Deliverables	Weeks
Phase 1 Requirements	Stakeholder interviews, questionnaire design and distribution, requirements elicitation, SRS document authoring in IEEE 830-1998 format, use case identification, Jira project setup	Approved SRS Document v1.0, Use Case List, Jira Project Structure	Week 2–3
Phase 2 System Design	Architecture design, Visual Paradigm UML diagrams (Use Case, Class, Sequence, Activity, ERD, State Machine), Figma UI/UX prototype design (all screens), database schema design, API endpoint specification	System Design Document, Complete Visual Paradigm Diagrams, Figma Prototype (all screens + flows)	Week 4–6
Phase 3 Implementation	Frontend development (React Native — all 4 subsystems), Backend API development (Node.js/Firebase), Firebase Firestore schema implementation, QR token engine, Cloud Functions, GitHub feature-branch workflow	Working Application (all Phase 1 features), GitHub Repository with documented commits	Week 7–10
Phase 4 Testing	Unit testing (API endpoints), Integration testing (subsystem interactions), User Acceptance Testing (UAT with sample UTM students), QR token security testing, performance testing, bug fixing	Test Plan Document, Test Results Report, Bug Tracker (Jira), UAT Sign-Off	Week 11–12
Phase 5 Deployment	Firebase Hosting deployment, App distribution via Expo EAS Build, final documentation, project presentation preparation, system demo recording	Deployed Application, Final Report, Presentation Slides, Demo Video	Week 13

[Insert Figure 9.1: Gantt Chart — Full Waterfall Phase Schedule with Week-by-Week Milestones]

Note: The Gantt chart above is produced and maintained in Jira. Weekly task assignments are visible in both the Jira timeline view and the team Trello board.

10.0 TEAM MEMBERS AND ROLES

10.1 Subsystem Ownership

Subsystem	Person In Charge	Matric No.
SS1 — User Management and Authentication Subsystem	[Member 1 Name]	A23CSXXXX
SS2 — Inventory and Listing Subsystem	[Member 2 Name]	A23CSXXXX
SS3 — Borrow Request and QR Handover Subsystem	[Member 3 Name]	A23CSXXXX
SS4 — Analytics and Administration Subsystem	[Member 4 Name]	A23CSXXXX

10.2 Waterfall Phase Roles

In the Waterfall methodology, team roles are defined per phase rather than rotating sprint-by-sprint. Each phase has a designated Phase Lead responsible for coordination, with all members contributing to implementation:

Waterfall Phase	Phase Lead	All Team Members Contribute To
Phase 1 — Requirements	[Member 1]	Questionnaire design, interview analysis, SRS authoring, use case writing
Phase 2 — System Design	[Member 2]	Visual Paradigm UML diagrams, Figma UI screens, database schema, API specification
Phase 3 — Implementation	[Member 3]	React Native frontend (per subsystem ownership), Node.js API, Firebase setup
Phase 4 — Testing	[Member 4]	Test case writing, unit/integration testing, UAT coordination, bug reporting
Phase 5 — Deployment	[Member 1]	Firebase deployment, documentation finalization, presentation preparation

10.3 Role Descriptions

Phase Lead: Coordinates all activities within the assigned Waterfall phase. Ensures all deliverables are completed to the required quality before phase sign-off. Reports phase status in weekly team meetings. Manages the corresponding Jira Epic and Trello board column.

Subsystem Developer: Responsible for implementing the assigned subsystem across all Waterfall phases — from requirements elicitation and design to implementation, testing, and deployment. Each developer owns their subsystem end-to-end.

Documentation Coordinator: Rotating across members. Responsible for maintaining the central project documentation repository in GitHub, ensuring all Visual Paradigm diagrams and Figma assets are version-controlled and up to date.

11.0 INDIVIDUAL SCHEDULE

Each team member's individual schedule reflects their subsystem ownership responsibilities across all five Waterfall phases. All individual schedules are maintained as Jira personal task boards and visualized in the Gantt charts below.

11.1 [Member 1 Name] — SS1: User Management and Authentication

Phase	Tasks	Week
Requirements	Elicit auth requirements, write FR-IT-001 to FR-IT-011, interview UTM IT dept re: OAuth2 API	Wk 2–3
Design	Draw Authentication Sequence Diagram, design Login/Profile screens in Figma, design users Firestore collection schema	Wk 4–6
Implementation	Build OAuth2 callback handler, JWT middleware, user profile CRUD, RBAC enforcement layer	Wk 7–10
Testing	Unit test auth endpoints, integration test SSO flow, UAT for login/logout/profile	Wk 11–12
Deployment	Configure Firebase Auth rules, finalize auth module documentation	Wk 13

[Insert Figure 11.1: Individual Gantt Chart — Member 1]

11.2 [Member 2 Name] — SS2: Inventory and Listing

Phase	Tasks	Week
Requirements	Elicit listing/search requirements, write FR-DI-001 to FR-DI-013, map UTM college geolocation data	Wk 2–3
Design	Draw Listing Class Diagram and Activity Diagram, design Item Listing and Search screens in Figma, design listings Firestore schema	Wk 4–6
Implementation	Build listing CRUD API, geolocation-filtered search endpoint, Firebase Storage photo upload, availability state management	Wk 7–10
Testing	Test search performance (target <2s), integration test listing states, UAT for create/search/browse	Wk 11–12
Deployment	Configure Firestore indexes for search queries, finalize inventory module documentation	Wk 13

[Insert Figure 11.2: Individual Gantt Chart — Member 2]

11.3 [Member 3 Name] — SS3: Borrow Request and QR Handover

Phase	Tasks	Week
Requirements	Elicit transaction requirements, write FR-BR, FR-QH, FR-LC requirements, design Lease Cycle State Machine	Wk 2–3

Design	Draw State Machine Diagram and full Sequence Diagrams for Happy Path, Error Path, Dispute Path in Visual Paradigm; design QR handover screens in Figma	Wk 4–6
Implementation	Build borrow request API, HMAC-SHA256 QR token engine (60s expiry), in-app camera scanner, Cloud Function state transitions, condition photo module	Wk 7–10
Testing	Security test QR replay attacks, integration test full transaction lifecycle, UAT for Happy Path and Conflict Path	Wk 11–12
Deployment	Deploy Cloud Functions, configure FCM notification triggers, finalize transaction module documentation	Wk 13

[Insert Figure 11.3: Individual Gantt Chart — Member 3]

11.4 [Member 4 Name] — SS4: Analytics and Administration

Phase	Tasks	Week
Requirements	Elicit admin/analytics requirements, write FR-IF, FR-NE, FR-AM requirements, define BI metrics with UTMCS stakeholder	Wk 2–3
Design	Design Admin Panel and Dashboard screens in Figma, draw Admin Use Case Diagram and audit log ERD in Visual Paradigm, define notification event schema	Wk 4–6
Implementation	Build Admin Panel (user management, dispute resolution), FCM notification engine, rating system, PDF export function, auditLogs Firestore collection	Wk 7–10
Testing	Test RBAC enforcement (admin vs user access), notification delivery timing, PDF export accuracy, UAT with admin personas	Wk 11–12
Deployment	Configure Firestore admin security rules, deploy PDF export function, finalize admin module documentation	Wk 13

[Insert Figure 11.4: Individual Gantt Chart — Member 4]

END OF PROJECT PROPOSAL

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